Top 10 Genes Ranked by Potency of Perturbation (Sorted by BMD Median)

DTTID: 108-020-004-000-5
Study Number: 108020004
Study Type: 5-Day Toxicity
Species/Strain: Rat HSD:Sprague Dawley SD

Test Compound: 1,4-Dichlorobenzene
CAS Number: 106-46-7
DTXSID: DTXSID1020431
Tissue: Kidney

Date: 22 May 2025
Time: 12:24:09 PM

Female

Gene Symbols	Entrez Gene IDs	Probe IDs	BMD_{rd25} ($BMDL_{rd25}$ - $BMDU_{rd25}$) in ppm	Maximum Fold Change	Direction of Expression Change
Srebf1	78968	SREBF1_32750	31.8 (7.8 - 123.9)	1.6	UP
Ccnb2	363088	CCNB2_8223	77.8 (29.5 - 183.5)	1.8	DOWN
Ccnb2-ps2	100364016	CCNB2_8223	77.8 (29.5 - 183.5)	1.8	DOWN
Cyp4a2	24306	CYP4A2_32530,CYP4A2_8426,CYP4A2_8425	125.1 (49.5 - 193.2)	3.1	UP
Cyp4a3	298423	CYP4A2_32530,CYP4A2_8426	137.6 (56.5 - 190.5)	2.9	UP
Kif22	293502	KIF22_8963	139.6 (48.2 - 291.1)	1.8	DOWN
Mki67	291234	MKI67_9232	150.4 (59.2 - 331.7)	1.7	DOWN
Cks2	498709	CKS2_8325	164.3 (55.0 - 1122.1)	1.6	DOWN
Dhcr7	64191	DHCR7_8466	190.4 (20.9 - NC)	1.5	DOWN
Mfap4	287382	MFAP4_32762	210.5 (55.7 - 447.5)	1.9	DOWN

rd25 = 25% relative deviation
NC = Non-convergent

Annotations:

Descriptions of orthologous human genes are shown due to the increased detail that is available in public resources such as UniprotKB (http://www.uniprot.org/uniprot/) and Entrez Gene (https://www.ncbi.nlm.nih.gov/gene/). Human UniprotKB was used as the primary resource due to the greater breadth of annotation and depth of functional detail that is provided. Rat UniprotKB was used as the second resource if the primary source did not provide a detailed description of function. Human Entrez gene summary was used as third resource. Rat Entrez gene summary was used as the fourth resource.

Gene definition version: https://doi.org/10.22427/NTP-DATA-002-00600-0002-000-0 V06132025

Srebf1: Human Uniprot function(Human SREBF1):[Sterol regulatory element-binding protein 1]: Precursor of the transcription factor form (Processed sterol regulatory element-binding protein 1), which is embedded in the endoplasmic reticulum membrane (PubMed:32322062). Low sterol concentrations promote processing of this form, releasing the transcription factor form that translocates into the nucleus and activates transcription of genes involved in cholesterol biosynthesis and lipid homeostasis (By similarity). \{ECO:0000250;UniProtKB:Q9WTN3, ECO:0000269;PubMed:32322062\}.; FUNCTION: [Processed sterol regulatory element-binding protein 1]: Key transcription factor that regulates expression of genes involved in cholesterol biosynthesis and lipid homeostasis (PubMed:8402897, PubMed:12177166, PubMed:32322062). Binds to the sterol regulatory element 1 (SRE-1) (5'-ATCACCCCAC-3'). Has dual sequence specificity binding to both an E-box motif (5'-ATCACGTGA-3') and to SRE-1 (5'-ATCACCCCAC-3') (PubMed:8402897, PubMed:12177166). Regulates the promoters of genes involved in cholesterol biosynthesis and the LDL receptor (LDLR) pathway of sterol regulation(PubMed:8402897, PubMed:12177166, PubMed:32322062). \{ECO:0000269;PubMed:12177166, ECO:0000269;PubMed:32322062, ECO:0000269;PubMed:8402897\}.; FUNCTION:[Isoform SREBP-1A]: Isoform expressed only in select tissues, which has higher transcriptional activity compared to SREBP-1C (By similarity). Able to stimulate both lipogenic and cholesterogenic gene expression(PubMed:12177166). Has a role in the nutritional regulation of fatty acids and triglycerides in lipogenic organs such as the liver (By similarity). Required for innate immune response in macrophages by regulating lipid metabolism (By similarity). \{ECO:0000250; UniProtKB:Q9WTN3, ECO:0000269;PubMed:12177166\}.; FUNCTION: [Isoform SREBP-1C]: Predominant isoform expressed in most tissues, which has weaker transcriptional activity compared to isoform SREBP-1A (By similarity). Primarily controls expression of lipogenic gene (PubMed:12177166). Strongly activates global lipid synthesis in rapidly growing cells (By similarity). \{ECO:0000250;UniProtKB:Q9WTN3, ECO:0000269;PubMed:12177166\}.; FUNCTION: [Isoform SREBP-1aDelta]: The absence of Golgi proteolytic processing requirement makes this isoform constitutively active in transactivation of lipogenic gene promoters. \{ECO:0000305;PubMed:7759101\}.; FUNCTION: [Isoform SREBP-1cDelta]: The absence of Golgi proteolytic processing requirement makes this isoform constitutively active in transactivation of lipogenic gene promoters. \{ECO:0000305;PubMed:7759101\}.

Ccnb2: Human Uniprot function (Human CCNB2): Essential for the control of the cell cycle at the G2/M (mitosis) transition.

Ccnb2-ps2: No description available.

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Cyp4a2: Human Uniprot function (Human CYP4A22): Catalyzes the omega- and (omega-1)-hydroxylation of various fatty acids such as laurate and palmitate. Shows no activity towards arachidonic acid and prostaglandin A1. Lacks functional activity in the kidney and does not contribute to renal 20-hydroxyeicosatetraenoic acid (20-HETE) biosynthesis. \{ECO:0000269;PubMed:10860550,ECO:0000269;PubMed:15611369\}.

Cyp4a3: Human Uniprot function (Human CYP4Z1): A cytochrome P450 monooxygenase that catalyzes omega and omega-1 hydroxylation of saturated fatty acids. Exhibits preferential omega versus omega-1 regioselectivity and (R) versus (S) stereoselectivity for hydroxylation of dodecanoic (lauric) acid. Mechanistically, uses molecular oxygen inserting one oxygen atom into a substrate, and reducing the second into a water molecule, with two electrons provided by NADPH via cytochrome P450 reductase (CPR; NADPH-ferrihemoprotein reductase). \{ECO:0000269;PubMed:10869363\}.

Kif22: Human Uniprot function (Human KIF22): Kinesin family member that is involved in spindle formation and the movements of chromosomes during mitosis and meiosis. Binds to microtubules and to DNA (By similarity). Plays a role in congression of laterally attached chromosomes in NDC80-depleted cells (PubMed:25743205). \{ECO:0000250;UniProtKB:Q9I869,ECO:0000269;PubMed:25743205\}.

Mki67: Human Uniprot function (Human MKI67): Required to maintain individual mitotic chromosomes dispersed in the cytoplasm following nuclear envelope disassembly (PubMed:27362226). Associates with the surface of the mitotic chromosome, the perichromosomal layer, and covers a substantial fraction of the chromosome surface (PubMed:27362226). Prevents chromosomes from collapsing into a single chromatin mass by forming a steric and electrostatic charge barrier: the protein has a high net electrical charge and acts as a surfactant, dispersing chromosomes and enabling independent chromosome motility (PubMed:27362226). Binds DNA, with a preference for supercoiled DNA and AT-rich DNA (PubMed:10878551). Does not contribute to the internal structure of mitotic chromosomes (By similarity). May play a role in chromatin organization (PubMed:24867636). It is however unclear whether it plays a direct role in chromatin organization or whether it is an indirect consequence of its function in maintaining mitotic chromosomes dispersed (Probable). \{ECO:0000250;UniProtKB:E9PVX6, ECO:0000269;PubMed:10878551, ECO:0000269;PubMed:24867636,ECO:0000269;PubMed:27362226\}.

Cks2: Human Uniprot function (Human CKS2): Binds to the catalytic subunit of the cyclin dependent kinases and is essential for their biological function.

Dhcr7: Human Uniprot function (Human DHCR7): 7-dehydrocholesterol reductase of the cholesterol biosynthetic pathway reducing the C7-C8 double bond of cholesta-5,7-dien-3beta-ol (7-dehydrocholesterol/7-DHC) and cholesta-5,7,24-trien-3beta-ol, two intermidiates in that pathway. {ECO:0000269|PubMed:25637936, ECO:0000269|PubMed:9465114, ECO:0000269|PubMed:9634533}.

Mfap4: Human Uniprot function (Human MFAP4): Could be involved in calcium-dependent cell adhesion or intercellular interactions. May contribute to the elastic fiber assembly and/or maintenance(PubMed:26601954). \{ECO:0000269;PubMed:26601954\}.

** END OF REPORT **